

COMICCRAFT – TEST REPORT

1. Introduction

The ComicCraft Test Report summarizes the testing performed on the ComicCraft AI Comic Story Creator application.

The purpose of testing is to verify that the application components work correctly and that the complete comic generation workflow operates as expected.

2. Testing Objective

The main objectives of testing are:

- Verify application functionality
- Identify errors and failures
- Verify communication between modules
- Check input validation
- Verify AI story generation
- Verify image processing
- Verify comic layout generation
- Verify PDF generation
- Check error handling

3. Modules Tested

The following modules were tested:

1. Frontend Interface
2. Story Prompt Input
3. FastAPI Backend
4. Gemini AI Integration
5. Story Panel Generation
6. Image Generation
7. Comic Layout
8. PDF Generation
9. API Communication
10. Error Handling

4. Testing Summary

A set of functional and integration test cases were performed.

The major test scenarios included:

- Launching the application
- Entering a valid story prompt
- Submitting an empty prompt
- Generating an AI story
- Creating comic panels
- Processing image prompts
- Generating panel images
- Creating the comic layout
- Generating a PDF

- Accessing the generated PDF
- Handling invalid requests
- Handling missing files

5. Test Results

The test results are summarized below:

Test Area	Expected Result	Test Result
Application Launch	Application opens successfully	Pass
Prompt Input	Prompt is accepted	Pass
Empty Prompt	Error message is displayed	Pass
Story Generation	AI generates story content	Pass
Panel Generation	Comic panels are created	Pass
Image Prompt	Image prompts are created	Pass
Image Processing	Panel images are processed	Pass
Comic Layout	Text and images are combined	Pass
PDF Generation	PDF is created	Pass
PDF Access	PDF is available	Pass
Error Handling	Errors are handled properly	Pass

6. Functional Testing Result

Functional testing verifies that the major features perform their intended operations.

The tested features include:

- Story prompt submission
- AI story generation
- Comic panel creation

- Image processing
- Comic layout creation
- PDF generation

The functional tests confirmed that the major application workflow operates according to the designed process.

7. Integration Testing Result

Integration testing verifies the communication between different modules.

The following integrations were checked:

Frontend → Backend

The frontend sends the user's story prompt to the FastAPI backend.

Backend → Gemini AI

The backend passes the validated prompt to the AI generation modules.

Story → Image Generation

The generated image prompts are passed to the image generation module.

Images → Comic Layout

Generated images are associated with their corresponding story panels.

Comic Layout → PDF

The completed comic layout is passed to the PDF generation module.

8. Error Handling Result

Error handling was tested for common problems such as:

- Empty prompts
- Invalid input
- Missing files
- Image generation failures
- PDF generation failures
- Invalid requests

The application is designed to return suitable error messages rather than failing without explanation.

9. Issues Identified During Development

During development and testing, issues may occur in areas such as:

- AI API configuration
- Image generation
- Missing image files
- PDF file generation
- File paths
- Invalid input
- API communication

These issues can be identified through testing and corrected during the development process.

10. Final Testing Status

The major functional areas of the ComicCraft application were tested.

The test process verifies the complete workflow:

User Prompt

↓

AI Story Generation

↓

Comic Panels

↓

Image Generation

↓

Comic Layout

↓

PDF Generation

↓

Final Comic

11. Conclusion

The testing process helps verify the functionality and integration of the ComicCraft application.

The Test Report provides a summary of the test activities, test results, module verification, and error handling.

Testing helps improve the reliability and usability of the final ComicCraft application.